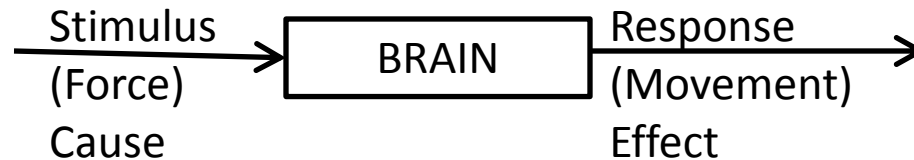


Cybernetics and Psychology

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Psyche as Mechanism



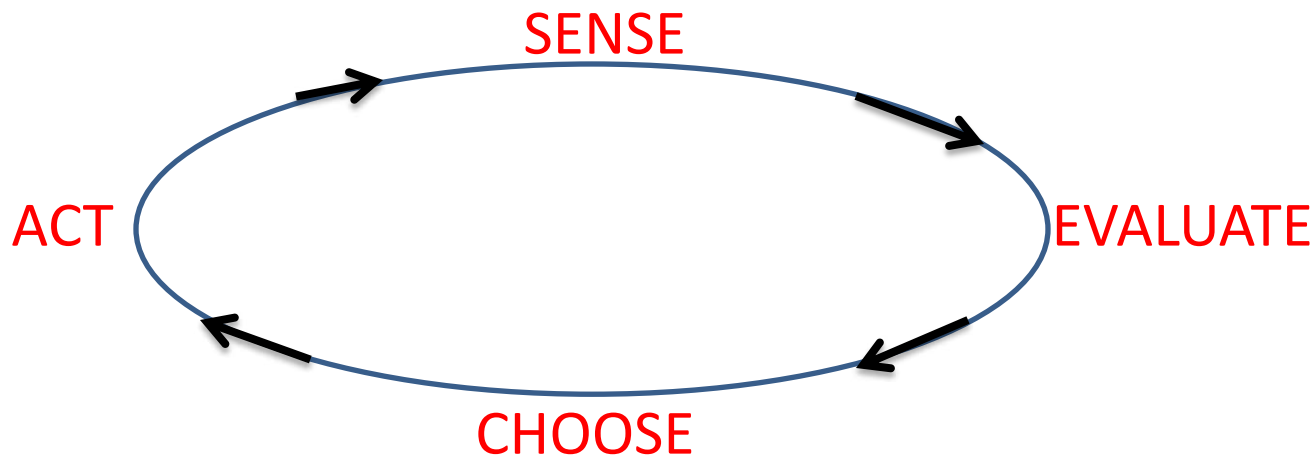
Mechanistic reductionism teaches that the universe consists only of matter and energy which are deterministically controlled by forces. Everything remains the same, until it is forced to change. Thus everything can be understood as a force causing a movement.

Behaviourism adopts mechanistic reductionism and argues that there is no such thing as mind, there is only a mental mechanism, a brain. Therefore all human behaviour can be understood as the stimulus (force) causing a response (movement) as an effect. Therefore human behaviour can be completely explained by the mechanism of stimulus and response.

The scientific method is a method for testing linear causal relationships. If the brain is merely a mechanism, it can be used to establish the formulae that will allow the scientist to predict human behaviour, and thus establish psychology as a science.

Psyche as a Control System

Cybernetics on the other hand views the human mind as a control system. A control system is a decision making process, and consists of four constituent processes.



Since these four processes form a feedback loop, a control process is non-linear. Therefore the control process is not an energy process, but a communication process. It processes information. Matter and energy are merely the media of communication and control.

A Cybernetic Control System is a Communication Process

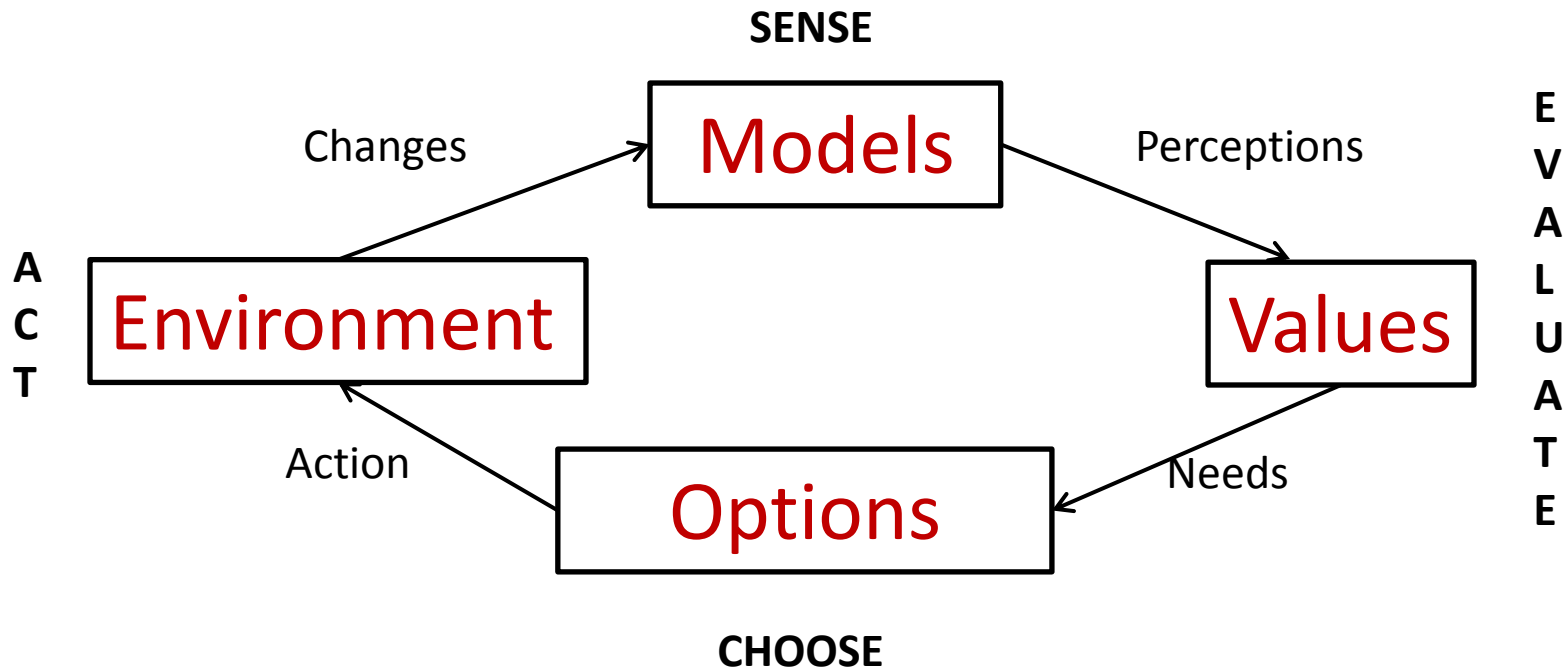
A communication process translates one form of information into another.

Sensing uses models to translate changes in the environment into perceptions.

Evaluating uses values to translate perceptions into evaluations.

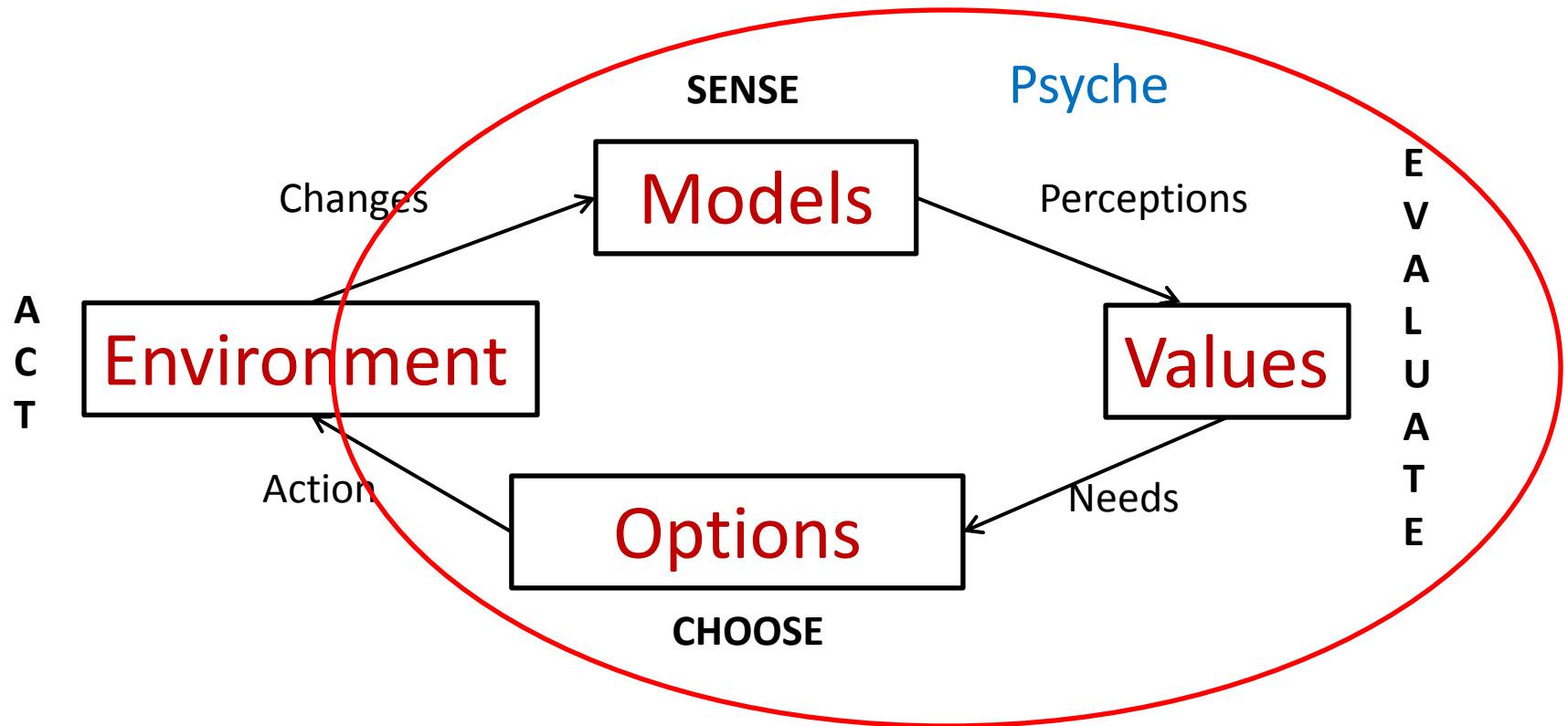
Choosing uses options to translate evaluations into action.

Acting consists of the environment translating actions into changes.



Models, values, options and environment are the translation processes that constitute the program that controls the control process.

The Psyche is the Internal Communication Process



The psyche is part of the environment, but it also seeks to change some aspect of the environment. Cybernetics sees action (behaviour) as purposeful. It seeks to create changes in the environment. It is creative.

Conscious and unconscious decision making

Unconscious decisions. Most of the decisions we make are unconscious. Our heart beat, our breathing, our digestion, our immune response. One of the first cybernetic studies, by Rosenbluth, was the way that the body uses feedback to make arm movements smooth.

Because information is categorical we also use feedback to learn to make distinctions and associations. This allows for more sophisticated and programmed responses. The most significant of these is learning language. We learn to distinguish between phonemes and then associate them into words, and sentences.

Consciousness. We can become aware of our sensory perception, our emotions, our actions, and our internal dialogue.

Self-conscious or reflexive decision-making takes the process to the meta-level. We can become aware of the programming as information and make decisions about the translators: exploring options, researching possible results, and defining criteria.

Problem Solving

Every problem is a decision waiting to be made.
What are the options expressed in the dilemma?

Options

Apply appropriate models.
What are the probable results of each option?

Models

Every decision is an exercise in values clarification.
What are the values implied by the options?

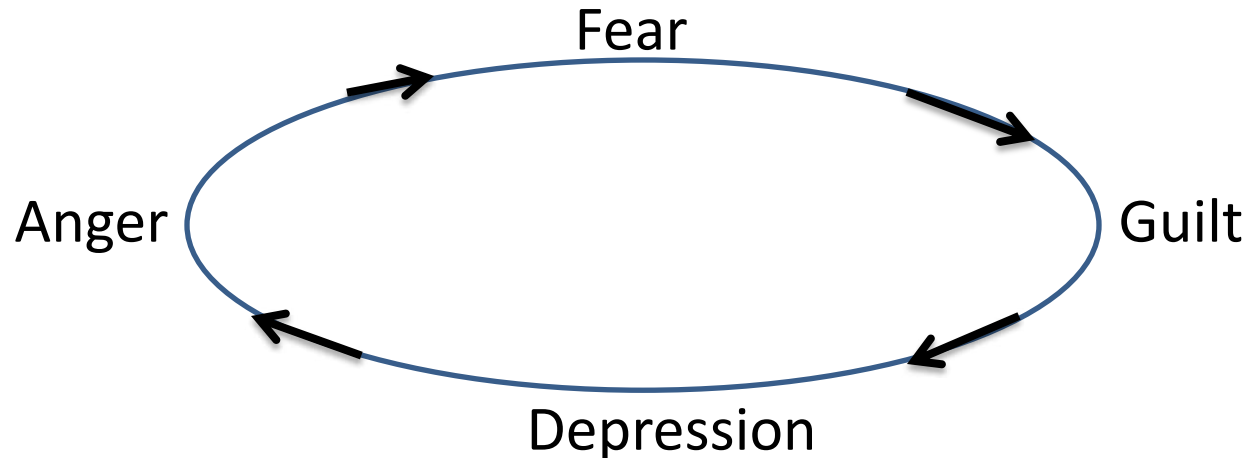
Values

A decision will emerge.

In relating effectively to our environment we seek **requisite variety** in our options (Skills, technology), realistic models (**Truth**), and functional values (**Ethics**).

The Negative Circle

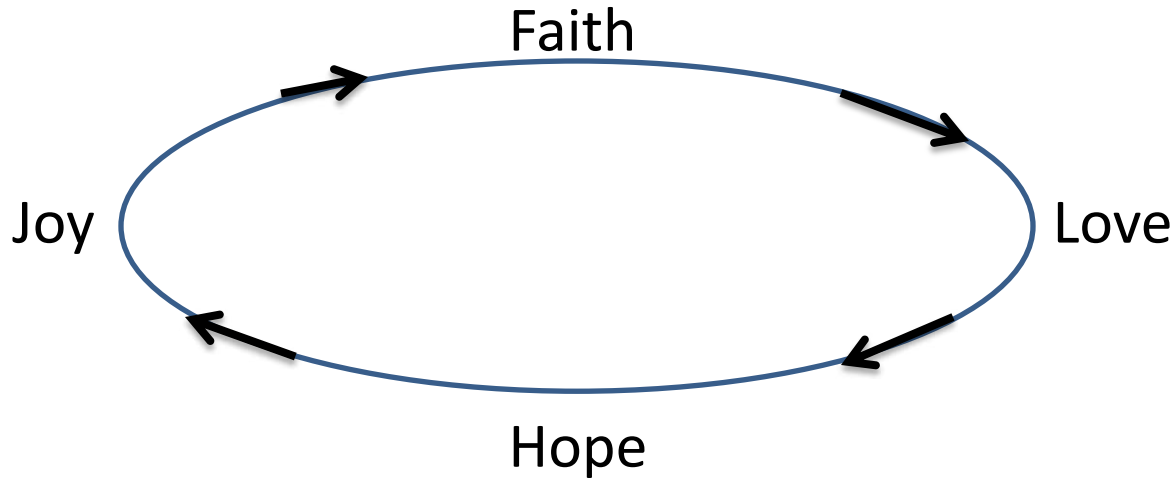
In working with families we found that the major struggle was with emotions. People were overwhelmed with feelings: fear, guilt, depression, anger.



We found that negative emotions formed a cycle. People would move from one negative emotion to another, as though it would solve the emotion. Fearful people would shift to blaming themselves. Guilty people would become depressed and hopeless to escape the guilt. Depressed people would get angry at others to relieve the depression. But living in a world of evil people is a very frightening thing. The negative emotions would just reinforced each other. Each person had their habitual spot.

The Positive Circle

Rather than move around the negative circle, it was necessary to move to a positive circle that would reinforce positive emotions.



Fear is a negative view of the world. Faith is a positive view of the world.

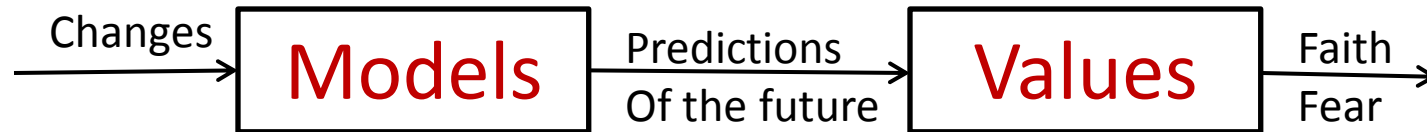
Guilt is blaming others. Love is forgiving others.

Depression is feeling trapped. Hope is having some possibilities.

Anger is a way of being in the world. Joy is the alternative way of being in the world.

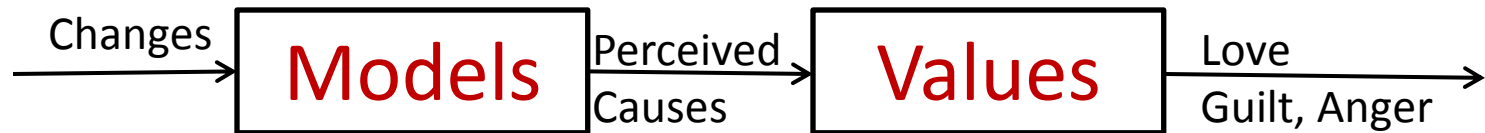
Emotions are the result of our models and our values.

Fear to faith



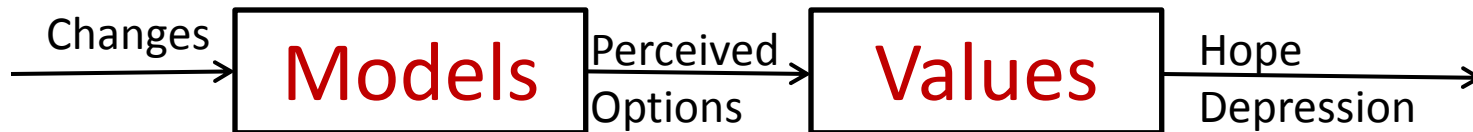
Our choice of models and values determine whether we look at the future with fear, or with faith.

Guilt and anger to love and forgiveness



Our choice of models and values determine whether we look at the causes with guilt and anger or love and forgiveness.

Depression to Hope



Our choice of models and values determine whether we look at the options with despair, or with hope.

We can look at the traditional psychiatric disorders as breakdowns in the translation processes.

Schizophrenia



Dysfunctional models will give us a distorted picture of our environment which can be a small distortion (climate change is a hoax) or an extreme distortion involving hallucinations, and delusions.

Addiction



Dysfunctional values will give us a distorted evaluation of our environment which can be a small obsession (phobias) or an extreme distortion involving an obsession or addiction.

Depression



Dysfunctional options will give us a distorted picture of our alternatives which can be a sense that there is only one option (compulsion) or a sense that there are no options (depression), or infinite options (mania).

These frequently work together as a syndrome. For example Obsessive Compulsive Disorder.

Model – I don't remember locking the door.

Perception ---- Did I lock the door?

Value – I should lock the door.

Need ---- I need to lock the door. (anxiety)

Option – My only option is to check. That will relieve my anxiety.

Action --- Check to see if it's locked.

There are a number of places to intervene in the syndrome.

A mnemonic device – take a picture of yourself locking the door.

Change the value – it would be nice to lock the door, but I don't have to.

Add an option – I can ignore the lock and enjoy the anxiety.

The pharmaceutical industry recommends drugs that will dampen the anxiety messages.

Cognitive Therapies

A great deal of work has been done on what are called cognitive or talk therapies that use communication to work on information processing programs.

Martin Seligman's Positive Psychology and his work on happiness.

David Burns and Feeding Good: The New Mood Therapy

Richard Bandler and John Grinder: NLP - Neuro Linguistic Programming

Communications Media

Matter and energy are the media of communication. The body uses a variety of chemicals, and electro-chemical process to act as media for biological information. The most famous of these is DNA which is a molecule with a valence that communicates genetic information, the genome.

The heart uses electrical impulses to control its rhythm. Surgeons can implant a pace maker that will augment this system when it malfunctions.

Researchers are constantly finding out more sophisticated chemical messaging systems such as the stress hormones, and the wake up hormones. The effort to produce chemicals that would force the body to feel a certain way has produced the popular drugs: heroin, meth-amphetamines, cocaine.

The End